DETAILED SECURITY REPORT · for security analysts and reviewers

test_flows.csv

Analysis #1 · report #1 · generated 2026-10-28 13:43 UTC · detector v3 · report schema 2.1

What did the detector predict, what evidence supports it, what remains uncertain, and what should an analyst check next?

1. Executive assessment

Offline analysis of 300 completed network flows uploaded as "test_flows.csv". The file contains flow measurements only: no live traffic, IP addresses, ports, timestamps or payloads.

Predicted Normal	Predicted attack	Needs review (confidence < 70%)	Mean confidence
34 (11.3%)	**266** (88.7%)	**151**	**73.0%**

Main findings

- 266 of 300 flows (88.7%) were predicted as attacks and 34 as Normal. The largest predicted groups are DoS (70), Exploits (45) and Generic (30). (model output)
- For the 70 DoS predictions, the most frequent reason was the average packet size sent by the source (a top-three reason in 62 of 70 flows). (XAI (TreeSHAP)) scope: group: predicted DoS
- For the 45 Exploits predictions, the most frequent reason was the number of recent connections to the same service and destination (a top-three reason in 37 of 45 flows). (XAI (TreeSHAP)) scope: group: predicted Exploits
- 151 predictions have confidence below 70%, and 133 predicted attacks could be more than one attack type. For 7 flows, Normal traffic is still a possible answer. (batch statistic)
- These are predictions about completed flows, not confirmed incidents or a forecast of future attacks. Check the flagged flows against logs before blocking anything. (proposed next step)

Same bullets as the Summary report. Written by code from the report record (counts, model outputs, TreeSHAP group findings); no LLM call. Each bullet states its origin and scope.

Prioritised next steps

- **A1** Review the low-confidence predictions — batch
- **A2** Check flow #284 in logs or captures — flow #284 (DoS)
- **A3** Check flow #134 in logs or captures — flow #134 (Exploits)

Batch triage priority: CRITICAL (rule: serious flows >= 25% of all flows, or malicious flows >= 50%). Batch triage priority: an application heuristic that orders review work from the predicted classes and fixed severity rules. It is not a model output, not observed business impact and not a confirmed compromise.

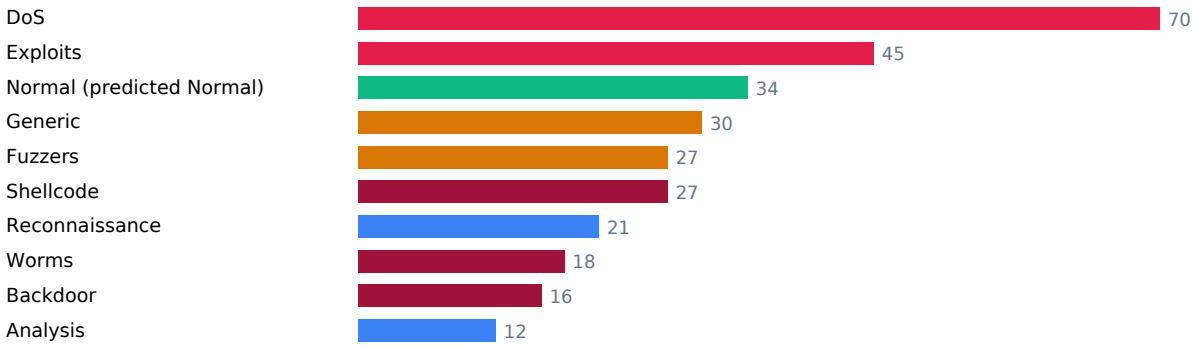
Most important uncertainty. 151 predictions have confidence below 70%.

Component status and coverage

Component	Status	Detail
Detection	**available**	300 flows analysed in 0.1 s
Explanations (TreeSHAP)	**available**	6 of 6 selected flows explained (TreeSHAP); LIME not generated
Group explanations	**available**	172 flows of 4 attack types explained for the group findings
Source retrieval (RAG)	**available**	32 distinct passages from 5 source families
Local LLM	**available**	generated: 5
Automated checks	**available**	24 generated statements checked, 0 rejected
PDF	**available**	

Explanation coverage: 6 illustrative flow cards, group findings over 172 flows; LLM explanations for 5 cards of 9 predicted attack types. Illustrative flows: for each of the four most frequent predicted attack types, the flow at the MEDIAN confidence of that type (a typical case, not the best one); the first flow of the review order (highest heuristic severity, then confidence); and the lowest-confidence flow whose shortlist contains Normal and an attack type (or, if there is none, the lowest-confidence attack-type-uncertain flow). Each card explains one flow, not its whole group.

2. Detection results and uncertainty



Predicted class	Flows	Share	Mean conf.	Severity (heuristic)	Status uncertain	Type uncertain
DoS	70	23.3%	56.1%	High	0	69
Exploits	45	15.0%	66.5%	High	0	25
Normal	34	11.3%	87.3%	-	7	0
Generic	30	10.0%	98.4%	Medium	0	0
Fuzzers	27	9.0%	71.4%	Medium	0	5
Shellcode	27	9.0%	71.8%	Critical	0	6
Reconnaissance	21	7.0%	97.1%	Low	0	1
Worms	18	6.0%	77.7%	Critical	0	7
Backdoor	16	5.3%	65.6%	Critical	0	8
Analysis	12	4.0%	58.8%	Low	0	12

Prediction sets (90% shortlists)

Shortlist	Flows
Single class (not a guarantee)	160
Several classes	140
Attack STATUS uncertain: Normal and an attack type	7
Attack TYPE uncertain: several attack types, no Normal	133
Predicted Normal, attack type also on the shortlist	9

- Probabilities are calibrated on held-out rows of the training file (temperature scaling). They describe the model, not proof, and can be less reliable on traffic that differs from the training data.
- The 90% prediction set (shortlist) targets 90% coverage on average over flows like the calibration rows; it is not a per-flow guarantee, and a single-class set is not certainty.
- Attack-STATUS uncertain: the shortlist contains Normal and at least one attack type (attack or not is unclear). Attack-TYPE uncertain: predicted attack and the shortlist has several attack types but not Normal.
- Confidence, heuristic severity and business impact are different quantities.
- On the official test file (shifted from the training file) the detector flagged 27.2% of Normal flows as attacks (model card).
- For the previous detector v2 the 90% sets contained the true class for 79.1% of official test flows (Notebook 21); this was not re-measured for v3.

Review order (heuristic severity, then confidence; not a list of confirmed threats)

Flow	Predicted class	Severity (heuristic)	Conf.	Shortlist	proto / service	Bytes src / dst
#172	Worms	Critical	99.4%	Worms	tcp / http	2,416 / 308,476
#18	Backdoor	Critical	98.7%	Backdoor	srp / -	200 / 0
#283	Worms	Critical	98.6%	Worms	tcp / http	3,114 / 482,584
#178	Worms	Critical	98.5%	Worms	tcp / http	2,314 / 308,456
#112	Worms	Critical	98.5%	Worms	tcp / http	1,302 / 268
#87	Backdoor	Critical	98.3%	Backdoor	etherip / -	200 / 0
#168	Worms	Critical	97.3%	Worms	tcp / http	1,282 / 268
#6	Worms	Critical	91.8%	Worms	tcp / http	1,296 / 268
#77	Backdoor	Critical	91.8%	Backdoor	fire / -	200 / 0
#64	Shellcode	Critical	91.7%	Shellcode	udp / -	196 / 0

3. Evidence and model explanation

Group findings describe all flows of a predicted type; finding cards describe ONE selected flow each. Contributions (TreeSHAP) explain how the detection model scored a flow, not what caused the traffic, and they are not shares of the probability. Full attribution tables and the raw evidence given to the LLM are in Appendix A and B.

Group findings (per predicted attack type)

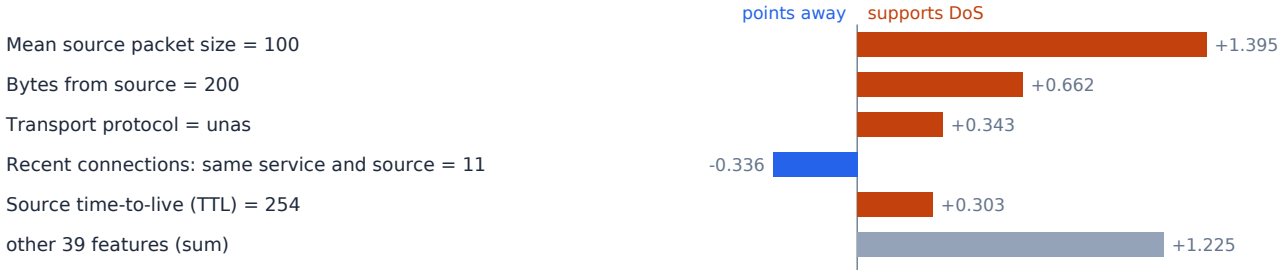
Predicted type	Flows explained	Most frequent supporting measurements (flows where it is a top-3 reason)	Most frequent opposing
DoS	70 of 70	Mean source packet size: 62 of 70 (median 100; between the 66th and 80th percentile of training flows (a very common value)) Bytes from source: 59 of 70 (median 200; between the 34th and 48th percentile of training flows (a very common value)) Transport protocol: 54 of 70 (most common value unas (46% of these flows); 6.9% of training flows have this value)	Recent connections: same service and source: 25 flows
Exploits	45 of 45	Recent connections: same service and destination: 37 of 45 (median 1; between the 0th and 25th percentile of training flows (a very common value)) Total bytes (engineered): 32 of 45 (median 1576; higher than about 66% of training flows) Source time-to-live (TTL): 21 of 45 (median 254; between the 35th and 100th percentile of training flows (a very common value))	Recent connections: same destination and source port: 22 flows
Generic	30 of 30	Application service: 29 of 30 (most common value dns (97% of these flows); 27.0% of training flows have this value) Total bytes (engineered): 29 of 30 (median 114; between the 4th and 27th percentile of training flows (a very common value)) Mean source packet size: 13 of 30 (median 57; between the 15th and 38th percentile of training flows (a very common value))	Recent connections: same source: 13 flows
Fuzzers	27 of 27	Source time-to-live (TTL): 27 of 27 (median 254; between the 35th and 100th percentile of training flows (a very common value)) Bytes from source: 23 of 27 (median 774; higher than about 61% of training flows) Application service: 11 of 27 (most common value - (89% of these flows); 53.6% of training flows have this value)	Recent connections: same source and destination: 6 flows

For each measurement: in how many flows it is one of the three strongest supporting contributions. Contributions are not added across flows. Values are summarised by median or most common value and compared with the training flows.

Finding F284 · flow #284: predicted DoS, probability 57.5%

Selected as: typical DoS flow: median confidence of the 70 flows predicted as DoS (rank 35 of 70 by confidence; one example, not the whole group).

Shortlist Analysis, Backdoor, DoS, Reconnaissance · Normal on shortlist: no · heuristic severity High · other classes: Exploits 15.9%, Analysis 12.0% (on shortlist), Backdoor 7.4% (on shortlist), Reconnaissance 4.9% (on shortlist)



Exact for the deployed score: base -4.760 + contributions = -1.170; ÷ temperature 1.159 + class offset +0.80 = -0.209.

Interpretation. For this flow, the strongest reasons towards DoS were mean source packet size = 100 and bytes from source = 200. Recent connections: same service and source = 11 pointed away from DoS. (XAI (TreeSHAP))

LLM explanation: accepted, wording in Appendix B · 5 statements checked, 0 rejected.

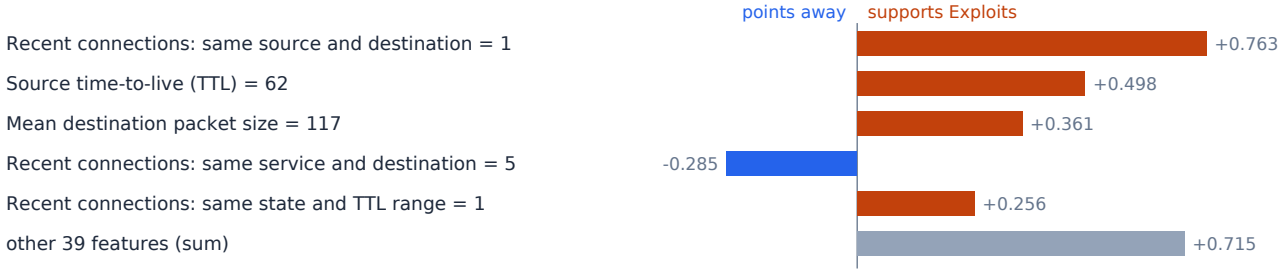
Alternatives and what is missing. Other classes the model did not rule out: Exploits 15.9%, Analysis 12.0% (on shortlist), Backdoor 7.4% (on shortlist), Reconnaissance 4.9% (on shortlist). Packet captures, sensor or host logs for this flow. The uploaded file has no addresses, ports or timestamps, so the flow has to be located by protocol, service, state and byte counts.

LIME: not generated. Full values and contributions: Appendix A (F284).

Finding F134 · flow #134: predicted Exploits, probability 66.0%

Selected as: typical Exploits flow: median confidence of the 45 flows predicted as Exploits (rank 23 of 45 by confidence; one example, not the whole group).

Shortlist Exploits · Normal on shortlist: no · heuristic severity High · other classes: Normal 17.4%, DoS 14.5%



Exact for the deployed score: base -3.721 + contributions = -1.412; ÷ temperature 1.159 + class offset +0.00 = -1.219.

Interpretation. For this flow, the strongest reasons towards Exploits were recent connections: same source and destination = 1 and source time-to-live (TTL) = 62. Recent connections: same service and destination = 5 pointed away from Exploits. (XAI (TreeSHAP))

LLM explanation: accepted, wording in Appendix B · 5 statements checked, 0 rejected.

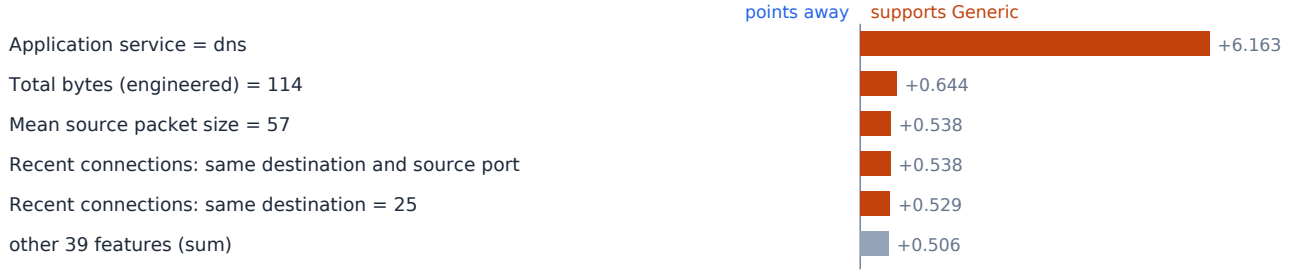
Alternatives and what is missing. Other classes the model did not rule out: Normal 17.4%, DoS 14.5%. Packet captures, sensor or host logs for this flow. The uploaded file has no addresses, ports or timestamps, so the flow has to be located by protocol, service, state and byte counts.

LIME: not generated. Full values and contributions: Appendix A (F134).

Finding F56 · flow #56: predicted Generic, probability above 99.5%

Selected as: typical Generic flow: median confidence of the 30 flows predicted as Generic (rank 15 of 30 by confidence; one example, not the whole group).

Shortlist Generic · Normal on shortlist: no · heuristic severity Medium



Exact for the deployed score: base -3.962 + contributions = +4.956; ÷ temperature 1.159 + class offset +0.20 = +4.476.

Interpretation. For this flow, the strongest reasons towards Generic were application service = dns and total bytes (engineered) = 114. None of the five strongest measurements pointed away from it. (XAI (TreeSHAP))

LLM explanation: accepted, wording in Appendix B · 5 statements checked, 0 rejected.

Alternatives and what is missing. No other class above 5%. Packet captures, sensor or host logs for this flow. The uploaded file has no addresses, ports or timestamps, so the flow has to be located by protocol, service, state and byte counts.

LIME: not generated. Full values and contributions: Appendix A (F56).

Finding F84 · flow #84: predicted Fuzzers, probability 71.8%

Selected as: typical Fuzzers flow: median confidence of the 27 flows predicted as Fuzzers (rank 14 of 27 by confidence; one example, not the whole group).

Shortlist Fuzzers · Normal on shortlist: no · heuristic severity Medium · other classes: Normal 27.0%



Exact for the deployed score: base -4.651 + contributions = +0.511; ÷ temperature 1.159 + class offset -0.20 = +0.241.

Interpretation. For this flow, the strongest reasons towards Fuzzers were bytes from source = 774 and source time-to-live (TTL) = 254. None of the five strongest measurements pointed away from it. (XAI (TreeSHAP))

LLM explanation: accepted, wording in Appendix B · 4 statements checked, 0 rejected.

Alternatives and what is missing. Other classes the model did not rule out: Normal 27.0%. Packet captures, sensor or host logs for this flow. The uploaded file has no addresses, ports or timestamps, so the flow has to be located by protocol, service, state and byte counts.

LIME: not generated. Full values and contributions: Appendix A (F84).

Finding F172 · flow #172: predicted Worms, probability 99.4%

Selected as: review priority: first flow of the review order (heuristic severity Critical, then confidence).

Shortlist Worms · Normal on shortlist: no · heuristic severity Critical



Exact for the deployed score: base -12.867 + contributions = +5.346; ÷ temperature 1.159 + class offset +0.00 = +4.612.

Interpretation. For this flow, the strongest reasons towards Worms were application service = http and source time-to-live (TTL) = 254. None of the five strongest measurements pointed away from it. (XAI (TreeSHAP))

LLM explanation: accepted, wording in Appendix B · 5 statements checked, 0 rejected.

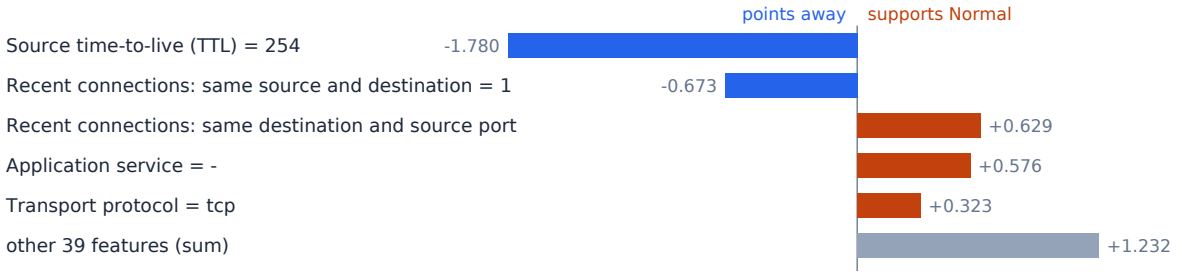
Alternatives and what is missing. No other class above 5%. Packet captures, sensor or host logs for this flow. The uploaded file has no addresses, ports or timestamps, so the flow has to be located by protocol, service, state and byte counts.

LIME: not generated. Full values and contributions: Appendix A (F172).

Finding F241 · flow #241: predicted Normal, probability 53.3%

Selected as: uncertain flow: the lowest-confidence flow whose 90% shortlist contains both Normal and an attack type (attack status uncertain).

Shortlist Fuzzers, Normal · Normal on shortlist: yes · heuristic severity None · other classes: Fuzzers 37.9% (on shortlist)



Exact for the deployed score: base -2.410 + contributions = -2.103; ÷ temperature 1.159 + class offset +0.00 = -1.814.

Interpretation. For this flow, the strongest reasons towards Normal were recent connections: same destination and source port = 1 and application service = -. Source time-to-live (TTL) = 254 pointed away from Normal. (XAI (TreeSHAP))

Alternatives and what is missing. Other classes the model did not rule out: Fuzzers 37.9% (on shortlist).

LIME: not generated. Full values and contributions: Appendix A (F241).

4. Retrieved context and source support

Retrieval selected passages from MITRE ATT&CK, NIST SP 800-61r3, NIST SP 800-94 and the CISA playbooks (Appendix C). Kinds of statement are labelled: **retrieved source** (quoted passage), **LLM, passed checks** (written from a cited passage; the check is word overlap, not meaning), and **heuristic** (the fixed class-to-technique mapping). A technique listed here is a hypothesis for context: a predicted class is not evidence that the technique was used (for example, Generic does not mean encryption was compromised, and Backdoor does not mean persistent access exists).

DoS — context for finding F284

Hypothesis (class mapping): T1498 Network Denial of Service, T1499 Endpoint Denial of Service (heuristic)

- T1498 Network Denial of Service [S1] (LLM, passed checks)
- Adversaries may perform Network Denial of Service (DoS) attacks to degrade or block the availability of targeted resources to users. [S1] (LLM, passed checks)

  Supporting sentence: “Adversaries may perform Network Denial of Service (DoS) attacks to degrade or block the availability of targeted resources to users.”

Exploits — context for finding F134

Hypothesis (class mapping): T1190 Exploit Public-Facing Application, T1210 Exploitation of Remote Services (heuristic)

- T1190 Exploit Public-Facing Application [S11] (LLM, passed checks)
- Adversaries may attempt to exploit a weakness in an Internet-facing host or system to initially access a network. [S11] (LLM, passed checks)
  Supporting sentence: “Adversaries may attempt to exploit a weakness in an Internet-facing host or system to initially access a network.”

Generic — context for finding F56

Hypothesis (class mapping): T1600 Weaken Encryption (heuristic)

- T1600 Weaken Encryption [S19] (LLM, passed checks)
- Adversaries may compromise a network device’s encryption capability in order to bypass encryption that would otherwise protect data communications. [S19] (LLM, passed checks)
  Supporting sentence: “Adversaries may compromise a network device’s encryption capability in order to bypass encryption that would otherwise protect data communications.”

Fuzzers — context for finding F84

Hypothesis (class mapping): T1595.002 Vulnerability Scanning, T1499.004 Application or System Exploitation (heuristic)

- T1595.002 Vulnerability Scanning [S21] (LLM, passed checks)
- Adversaries may scan victims for vulnerabilities that can be used during targeting. Vulnerability scans typically check if the configuration of a target host/application (ex: software and version) potentially aligns with the target of a specific exploit the adversary may seek to use. [S21] (LLM, passed checks)
  Supporting sentence: “Vulnerability scans typically check if the configuration of a target host/application (ex: software and version) potentially aligns with the target of a specific exploit the adversary may seek to use.”

Worms — context for finding F172

Hypothesis (class mapping): T1210 Exploitation of Remote Services, T1570 Lateral Tool Transfer (heuristic)

- T1210 Exploitation of Remote Services [S12] (LLM, passed checks)
- Adversaries may exploit remote services to gain unauthorized access to internal systems once inside of a network. [S12] (LLM, passed checks)
  Supporting sentence: “Adversaries may exploit remote services to gain unauthorized access to internal systems once inside of a network.”

5. Response and mitigation plan

The application never executes these actions; the checklist only records analyst decisions. Containment steps need the evidence named under "Requires"; a predicted class alone never justifies blocking, isolation or credential changes. General prevention is listed separately from incident response.

Verify now (read-only)

A1. Review the low-confidence predictions · batch · proposed next step

Open the flow list filtered to confidence Low and review the 151 flows below 70% before acting on any of them.

Why: Low-confidence predictions are the most likely to be wrong.

Requires: Always; read-only. · **Owner:** Security analyst · **Possible disruption:** None (read-only). · **Check:** Each reviewed flow is marked confirmed, false positive or unclear in the checklist.

Id	Flow to locate in captures or logs	Why (model output)	Evidence
A2	Find flow #284 (unas/-, state INT, 200 bytes sent, 0 received) in packet captures or sensor logs from the same period and decide whether it is DoS activity.	The model predicts DoS with calibrated probability 0.58; its 90% shortlist is Analysis, Backdoor, DoS, Reconnaissance. A prediction is not a confirmed incident.	[F284.E1] [F284.E2] [S1] [S31]
A3	Find flow #134 (tcp/-, state FIN, 836 bytes sent, 1166 received) in packet captures or sensor logs from the same period and decide whether it is Exploits activity.	The model predicts Exploits with calibrated probability 0.66; its 90% shortlist is Exploits. A prediction is not a confirmed incident.	[F134.E1] [F134.E2] [S11] [S31]

Id	Flow to locate in captures or logs	Why (model output)	Evidence
A4	Find flow #56 (udp/dns, state INT, 114 bytes sent, 0 received) in packet captures or sensor logs from the same period and decide whether it is Generic activity.	The model predicts Generic with calibrated probability above 0.99; its 90% shortlist is Generic. A prediction is not a confirmed incident.	[F56.E1] [F56.E2] [S19] [S31]
A5	Find flow #84 (tcp/-, state FIN, 774 bytes sent, 642 received) in packet captures or sensor logs from the same period and decide whether it is Fuzzers activity.	The model predicts Fuzzers with calibrated probability 0.72; its 90% shortlist is Fuzzers. A prediction is not a confirmed incident.	[F84.E1] [F84.E2] [S21] [S31]
A6	Find flow #172 (tcp/http, state FIN, 2416 bytes sent, 308476 received) in packet captures or sensor logs from the same period and decide whether it is Worms activity.	The model predicts Worms with calibrated probability 0.99; its 90% shortlist is Worms. A prediction is not a confirmed incident.	[F172.E1] [F172.E2] [S12] [S31]

For every flow above. Requires: Always first; read-only. Nothing should be blocked on the prediction alone. Owner: Security analyst. Disruption: None (read-only). Confirmed if logs or captures show activity matching the cited description of the predicted type; weakened if the traffic belongs to a known service or job of your own team, other sensors show nothing for that period, or Normal is on the flow's shortlist.

Contain only if confirmed

A7. Filter or block the confirmed traffic · attack type DoS · LLM, passed checks

Filter Network Traffic: When flood volumes exceed the capacity of the network connection being targeted, it is typically necessary to intercept the incoming traffic upstream to filter out the attack traffic from the legitimate traffic. [S7]

Why: Guidance retrieved for the DoS prediction (see the cited passage). It applies only after verification.

Requires: Packet captures or sensor logs confirm the traffic is malicious AND its source, destination and service are identified (the uploaded file has none of these). A DoS prediction alone is not enough. · **Owner:** Network engineering · **Possible disruption:** Can also block legitimate users of the same service or address range. · **Check:** Sensor counts for the matching traffic drop while service health checks stay normal. · **Undo:** Remove the rule or let it expire; record the rule and who approved it.

A8. Filter or block the confirmed traffic · attack type Exploits · LLM, passed checks

Filter Network Traffic: Restrict outbound network traffic from public-facing servers to prevent unauthorized connections from initiating communications with attacker-controlled infrastructure. [S16]

Why: Guidance retrieved for the Exploits prediction (see the cited passage). It applies only after verification.

Requires: Packet captures or sensor logs confirm the traffic is malicious AND its source, destination and service are identified (the uploaded file has none of these). A Exploits prediction alone is not enough. · **Owner:** Network engineering · **Possible disruption:** Can also block legitimate users of the same service or address range. · **Check:** Sensor counts for the matching traffic drop while service health checks stay normal. · **Undo:** Remove the rule or let it expire; record the rule and who approved it.

A9. Reset exposed credentials or keys · attack type Generic · LLM, passed checks

Change system admin passwords, rotate private keys and service/application account secrets where compromise is suspected. [S18]

Why: Guidance retrieved for the Generic prediction (see the cited passage). It applies only after verification.

Requires: Authentication logs or endpoint alerts show that a named account, password or key was exposed or misused. Network-flow predictions cannot show this. A Generic prediction alone is not enough. · **Owner:** System administrator / identity team · **Possible disruption:** Users and services must sign in again; jobs using the old secret fail until updated. · **Check:** The old credentials are rejected and no new suspicious sign-ins appear. · **Undo:** Not reversible: update dependent services before rotating.

A10. Filter or block the confirmed traffic · attack type Generic · LLM, passed checks

Perform blocking (and logging) of unauthorized accesses, malware sources, and egress traffic to known attacker Internet Protocol (IP) addresses. [S18]

Why: Guidance retrieved for the Generic prediction (see the cited passage). It applies only after verification.

Requires: Packet captures or sensor logs confirm the traffic is malicious AND its source, destination and service are identified (the uploaded file has none of these). A Generic prediction alone is not enough. · **Owner:** Network engineering · **Possible disruption:** Can also block legitimate users of the same service or address range. · **Check:** Sensor counts for the matching traffic drop while service health checks stay normal. · **Undo:** Remove the rule or let it expire; record the rule and who approved it.

Recover and monitor

A11. Restore and monitor after a confirmed incident · batch · retrieved source

See NIST SP 800-184: Guide for Cybersecurity Event Recovery for additional guidance. - Developing response scenarios for threat actor use of alternative attack vectors. - Allowing adequate time to ensure all systems are clear of all possible threat actor persistence mechanisms (backdoors, etc.) as adversaries often use more than one mechanism. ... [S32]

Why: Recovery guidance retrieved for a confirmed incident.

Requires: Only after a confirmed incident has been contained. · **Owner:** Incident response lead with the system owners · **Possible disruption:** Restoring systems may need planned downtime. · **Check:** Restored systems behave normally and monitoring shows no recurrence for an agreed period.

General prevention (separate from incident response)

- **attack type DoS:** Ensure that unnecessary ports and services are closed to prevent risk of discovery and potential exploitation. [S6]
- **attack type Exploits:** Application Isolation and Sandboxing: Application isolation will limit what other processes and system features the exploited target can access. [S14]
- **attack type Fuzzers:** Efforts should focus on minimizing the amount and sensitivity of data available to external parties. [S24]
- **attack type Worms:** Regularly scan the internal network for available services to identify new and potentially vulnerable services. [S15]
- **attack type Worms:** Disable or Remove Feature or Program: Minimize available services to only those that are necessary. [S13]

6. Limitations and provenance

What is not known

- No IP addresses, ports, host names, timestamps or payloads are in the uploaded file, so hosts, timelines and botnet or DNS attribution cannot be established.
- Predictions are not confirmed incidents; class-to-ATT&CK mappings are hypotheses for context.
- Heuristic severity and triage priority are application rules, not measured impact.
- Illustrative flows cover 6 of 300 flows; other flows of the same type may differ.

Automated checks

rule-based: citation validity, typed fact binding (values, contributions, signs, probabilities, prediction set), class / column / number checks, no value judgements or causal claims, lexical overlap with cited passages, real ATT&CK IDs present in cited passages. Not a semantic proof.

Human review: not reviewed yet (see the action checklist).

Versions: detector LightGBM v3 (class offsets, chosen without test labels) (v3); report schema 2.1, template summary + detailed 2.1; knowledge base downloaded 2026-10-18; LLM qwen2.5:7b (local); input file sha256 e8f8eb325c36dd3c…. Full hashes: Appendix D.

Appendix A. Measured values and full attributions

Values observed in the upload, their position among the training flows (training rows only), and every TreeSHAP contribution of the predicted class, largest first.

F284 · flow #284 (DoS)

Measurement	Value	Compared with training flows
Transport protocol (proto)	unas	6.9% of training flows have this value
Application service (service)	-	53.6% of training flows have this value
Connection state (state)	INT	46.8% of training flows have this value
Flow duration (dur)	0.000011	higher than about 45% of training flows
Bytes from source (sbytes)	200	between the 34th and 48th percentile of training flows (a very common value)
Bytes from destination (dbytes)	0	between the 0th and 48th percentile of training flows (a very common value)
Packets from source (spkts)	2	between the 2nd and 51st percentile of training flows (a very common value)
Packets from destination (dpkts)	0	between the 0th and 48th percentile of training flows (a very common value)
Source time-to-live (TTL) (sttl)	254	between the 35th and 100th percentile of training flows (a very common value)
Destination time-to-live (TTL) (dttl)	0	between the 0th and 48th percentile of training flows (a very common value)
Mean source packet size (smean)	100	between the 66th and 80th percentile of training flows (a very common value)
Mean destination packet size (dmean)	0	between the 0th and 48th percentile of training flows (a very common value)

Feature	Value	Contrib.	Feature	Value	Contrib.
smean	100	+1.3950	synack	0	+0.0683
sbytes	200	+0.6618	dtcpb	0	+0.0672
proto	unas	+0.3426	service	-	-0.0629
ct_srv_src	11	-0.3364	djit	0	+0.0600
sttl	254	+0.3027	stcpb	0	+0.0569
fe_total_bytes	200	+0.1918	sjit	0	+0.0359
tcprtt	0	+0.1560	dbytes	0	-0.0321
ct_src_ltm	5	+0.1437	ct_dst_src_ltm	2	+0.0311
ct_dst_ltm	2	+0.1195	dloss	0	+0.0302
rate	90909.0902	+0.0937	dur	0.000011	+0.0222
fe_byte_balance	1	+0.0890	sload	72727272	+0.0193
ct_srv_dst	11	+0.0704	swin	0	+0.0182

Largest 24 of 44 contributions shown; the full list is in the JSON export.

F134 · flow #134 (Exploits)

Measurement	Value	Compared with training flows
Transport protocol (proto)	tcp	45.6% of training flows have this value
Application service (service)	-	53.6% of training flows have this value
Connection state (state)	FIN	44.4% of training flows have this value
Flow duration (dur)	0.68	higher than about 76% of training flows
Bytes from source (sbytes)	836	higher than about 64% of training flows
Bytes from destination (dbytes)	1166	higher than about 76% of training flows

Measurement	Value	Compared with training flows
Packets from source (spkts)	10	between the 56th and 72nd percentile of training flows (a very common value)
Packets from destination (dpkts)	10	between the 73rd and 77th percentile of training flows (a very common value)
Source time-to-live (TTL) (sttl)	62	between the 25th and 35th percentile of training flows (a very common value)
Destination time-to-live (TTL) (dttl)	252	between the 72nd and 100th percentile of training flows (a very common value)
Mean source packet size (smean)	84	higher than about 57% of training flows
Mean destination packet size (dmean)	117	higher than about 82% of training flows

Feature	Value	Contrib.	Feature	Value	Contrib.
ct_dst_src_ltm	1	+0.7633	sbytes	836	+0.1207
sttl	62	+0.4983	ct_dst_sport_ltm	1	-0.1048
dmean	117	+0.3607	ct_src_ltm	2	+0.1011
ct_srv_dst	5	-0.2852	ct_src_dport_ltm	1	+0.0941
ct_state_ttl	1	+0.2564	dtcpb	3933965920	-0.0888
dttl	252	+0.2229	tcprtt	0.121	-0.0758
proto	tcp	+0.2069	dload	12352.4688	+0.0712
ct_srv_src	10	-0.1898	service	-	-0.0537
smean	84	+0.1826	sloss	2	+0.0445
fe_total_bytes	2002	+0.1380	rate	27.9401	+0.0404
dbytes	1166	-0.1340	spkts	10	-0.0340
sjit	4759.5295	+0.1208	sinpkt	75.5584	+0.0328

Largest 24 of 44 contributions shown; the full list is in the JSON export.

F56 · flow #56 (Generic)

Measurement	Value	Compared with training flows
Transport protocol (proto)	udp	36.1% of training flows have this value
Application service (service)	dns	27.0% of training flows have this value
Connection state (state)	INT	46.8% of training flows have this value
Flow duration (dur)	0.000026	higher than about 46% of training flows
Bytes from source (sbytes)	114	between the 5th and 27th percentile of training flows (a very common value)
Bytes from destination (dbytes)	0	between the 0th and 48th percentile of training flows (a very common value)
Packets from source (spkts)	2	between the 2nd and 51st percentile of training flows (a very common value)
Packets from destination (dpkts)	0	between the 0th and 48th percentile of training flows (a very common value)
Source time-to-live (TTL) (sttl)	254	between the 35th and 100th percentile of training flows (a very common value)
Destination time-to-live (TTL) (dttl)	0	between the 0th and 48th percentile of training flows (a very common value)
Mean source packet size (smean)	57	between the 15th and 38th percentile of training flows (a very common value)
Mean destination packet size (dmean)	0	between the 0th and 48th percentile of training flows (a very common value)

Feature	Value	Contrib.	Feature	Value	Contrib.
service	dns	+6.1633	dload	0	+0.1412

Feature	Value	Contrib.	Feature	Value	Contrib.
fe_total_bytes	114	+0.6441	dbytes	0	+0.0970
smean	57	+0.5381	rate	38461.5389	-0.0875
ct_dst_sport_ltm	25	+0.5377	stcpb	0	+0.0710
ct_dst_ltm	25	+0.5285	fe_byte_balance	1	+0.0628
ct_srv_src	45	+0.4393	djit	0	+0.0540
dur	0.000026	-0.3048	dtcpb	0	+0.0498
sttl	254	+0.2078	dloss	0	-0.0483
ct_dst_src_ltm	45	+0.1745	sbytes	114	+0.0453
ct_src_ltm	25	-0.1719	dinpkt	0	+0.0439
ct_src_dport_ltm	25	-0.1576	proto	udp	-0.0390
ct_srv_dst	45	-0.1501	sload	17538462	+0.0364

Largest 24 of 44 contributions shown; the full list is in the JSON export.

F84 · flow #84 (Fuzzers)

Measurement	Value	Compared with training flows
Transport protocol (proto)	tcp	45.6% of training flows have this value
Application service (service)	-	53.6% of training flows have this value
Connection state (state)	FIN	44.4% of training flows have this value
Flow duration (dur)	0.8067	higher than about 79% of training flows
Bytes from source (sbytes)	774	higher than about 61% of training flows
Bytes from destination (dbytes)	642	higher than about 70% of training flows
Packets from source (spkts)	14	higher than about 76% of training flows
Packets from destination (dpkts)	12	higher than about 77% of training flows
Source time-to-live (TTL) (sttl)	254	between the 35th and 100th percentile of training flows (a very common value)
Destination time-to-live (TTL) (dttl)	252	between the 72nd and 100th percentile of training flows (a very common value)
Mean source packet size (smean)	55	higher than about 12% of training flows
Mean destination packet size (dmean)	54	higher than about 64% of training flows

Feature	Value	Contrib.	Feature	Value	Contrib.
sbytes	774	+1.4870	smean	55	+0.1006
sttl	254	+1.0318	ct_state_ttl	1	+0.0956
dbytes	642	+0.6119	dloss	3	-0.0948
service	-	+0.4067	ct_src_ltm	2	+0.0897
synack	0.0678	+0.2508	fe_total_bytes	1416	-0.0856
fe_byte_balance	0.0932	+0.2463	dinpkt	67.1724	+0.0799
ct_dst_sport_ltm	1	+0.1904	djit	107.2373	-0.0633
ct_dst_src_ltm	2	+0.1756	proto	tcp	+0.0398
ct_srv_dst	5	+0.1555	dpkts	12	+0.0322
dmean	54	+0.1337	dur	0.8067	+0.0191
ct_srv_src	7	+0.1199	ackdat	0.0669	+0.0189
ct_src_dport_ltm	2	+0.1102	dtcpb	2870062246	+0.0157

Largest 24 of 44 contributions shown; the full list is in the JSON export.

F172 · flow #172 (Worms)

Measurement	Value	Compared with training flows
Transport protocol (proto)	tcp	45.6% of training flows have this value
Application service (service)	http	10.7% of training flows have this value
Connection state (state)	FIN	44.4% of training flows have this value
Flow duration (dur)	3.9937	higher than about 96% of training flows
Bytes from source (sbytes)	2416	higher than about 83% of training flows
Bytes from destination (dbytes)	308476	higher than about 99% of training flows
Packets from source (spkts)	48	higher than about 90% of training flows
Packets from destination (dpkts)	236	higher than about 99% of training flows
Source time-to-live (TTL) (sttl)	254	between the 35th and 100th percentile of training flows (a very common value)
Destination time-to-live (TTL) (dttl)	252	between the 72nd and 100th percentile of training flows (a very common value)
Mean source packet size (smean)	50	higher than about 6% of training flows
Mean destination packet size (dmean)	1307	higher than about 100% of training flows

Feature	Value	Contrib.	Feature	Value	Contrib.
service	http	+3.4728	stcpb	2507864534	+0.2605
sttl	254	+3.1488	ct_dst_src_ltm	1	+0.2472
sbytes	2416	+3.0094	dur	3.9937	+0.2414
smean	50	+1.7735	rate	70.8607	+0.2413
fe_byte_balance	-0.9845	+1.5665	sloss	2	+0.2237
ct_srv_src	1	+0.6592	sinpkt	84.9734	+0.2191
synack	0.1012	+0.5937	sload	4739.4053	+0.1980
dinpkt	16.5611	+0.4862	ct_dst_ltm	1	+0.1632
ackdat	0.1165	+0.4116	dmean	1307	+0.1528
ct_srv_dst	2	+0.4061	tcprtt	0.2177	+0.1453
sjit	4002.9769	+0.3228	ct_src_ltm	1	+0.1295
dload	615299.375	+0.2679	fe_total_bytes	310892	-0.0587

Largest 24 of 44 contributions shown; the full list is in the JSON export.

F241 · flow #241 (Normal)

Measurement	Value	Compared with training flows
Transport protocol (proto)	tcp	45.6% of training flows have this value
Application service (service)	-	53.6% of training flows have this value
Connection state (state)	FIN	44.4% of training flows have this value
Flow duration (dur)	1.0064	higher than about 83% of training flows
Bytes from source (sbytes)	534	higher than about 52% of training flows
Bytes from destination (dbytes)	268	between the 54th and 60th percentile of training flows (a very common value)
Packets from source (spkts)	10	between the 56th and 72nd percentile of training flows (a very common value)
Packets from destination (dpkts)	6	between the 56th and 63rd percentile of training flows (a very common value)
Source time-to-live (TTL) (sttl)	254	between the 35th and 100th percentile of training flows (a very common value)
Destination time-to-live (TTL) (dttl)	252	between the 72nd and 100th percentile of training flows (a very common value)
Mean source packet size (smean)	53	higher than about 8% of training flows

Measurement	Value	Compared with training flows
Mean destination packet size (dmean)	45	between the 56th and 63rd percentile of training flows (a very common value)

Feature	Value	Contrib.	Feature	Value	Contrib.
sttl	254	-1.7796	ct_dst_ltm	1	+0.0983
ct_dst_src_ltm	1	-0.6730	ct_srv_src	6	+0.0908
ct_dst_sport_ltm	1	+0.6292	djit	289.3137	+0.0712
service	-	+0.5760	ct_srv_dst	3	+0.0708
proto	tcp	+0.3228	state	FIN	+0.0636
dttl	252	-0.3008	synack	0.1327	+0.0549
smean	53	+0.3005	dloss	1	+0.0520
sbytes	534	+0.2043	tcprtt	0.2053	+0.0500
fe_total_bytes	802	+0.1772	dbytes	268	+0.0451
swin	255	+0.1664	dmean	45	+0.0432
dinpkt	174.7448	+0.1207	trans_depth	0	+0.0421
ct_state_ttl	1	-0.1146	fe_byte_balance	0.3317	+0.0414

Largest 24 of 44 contributions shown; the full list is in the JSON export.

Appendix B. Evidence given to the LLM and rejected statements

F284 (DoS)

Evidence id	Fact (computed by code)
F284.E1	Predicted class: DoS with calibrated probability 0.58
F284.E2	Conformal prediction set (90% coverage): Analysis, Backdoor, DoS, Reconnaissance
F284.E3	Column smean = 100 supports DoS (contribution +1.395)
F284.E4	Column sbytes = 200 supports DoS (contribution +0.662)
F284.E5	Column proto = unas supports DoS (contribution +0.343)
F284.E6	Column ct_srv_src = 11 argues against DoS (contribution -0.336)
F284.E7	Column sttl = 254 supports DoS (contribution +0.303)
F284.E8	In the analysed dataset, 70 flows (26.3% of the predicted attacks) were classified as DoS, with average confidence 0.56

LLM explanation (passed checks). The model is 0.58 confident that this flow is DoS with a 90% prediction set containing Analysis, Backdoor, DoS, and Reconnaissance. The strongest evidence is smean = 100 (contribution +1.395), sbytes = 200 (contribution +0.662), and proto = unas (contribution +0.343), while ct_srv_src = 11 (contribution -0.336) argues against it. In the analysed dataset, 70 flows (26.3% of the predicted attacks) were classified as DoS, with average confidence 0.56. [F284.E1] [F284.E2] [F284.E3] [F284.E4] [F284.E5] [F284.E6] [F284.E7] [F284.E8]

Generation: status generated, model qwen2.5:7b, 8 s, 5 statements checked, 0 rejected.

F134 (Exploits)

Evidence id	Fact (computed by code)
F134.E1	Predicted class: Exploits with calibrated probability 0.66
F134.E2	Conformal prediction set (90% coverage): Exploits
F134.E3	Column ct_dst_src_ltm = 1 supports Exploits (contribution +0.763)
F134.E4	Column sttl = 62 supports Exploits (contribution +0.498)
F134.E5	Column dmean = 117 supports Exploits (contribution +0.361)
F134.E6	Column ct_srv_dst = 5 argues against Exploits (contribution -0.285)

Evidence id	Fact (computed by code)
F134.E7	Column ct_state_ttl = 1 supports Exploits (contribution +0.256)
F134.E8	In the analysed dataset, 45 flows (16.9% of the predicted attacks) were classified as Exploits, with average confidence 0.67

LLM explanation (passed checks). The model is 0.66 confident that this flow is Exploits, and its 90% prediction set contains Exploits. The strongest evidence is ct_dst_src_ltm = 1 (contribution +0.763), sttl = 62 (contribution +0.498), and dmean = 117 (contribution +0.361), while ct_srv_dst = 5 (contribution -0.285) argues against it. In the analysed dataset, 45 flows (16.9% of the predicted attacks) were classified as Exploits, with average confidence 0.67. [F134.E1] [F134.E2] [F134.E3] [F134.E4] [F134.E5] [F134.E6] [F134.E8]

Generation: status generated, model qwen2.5:7b, 8 s, 5 statements checked, 0 rejected.

F56 (Generic)

Evidence id	Fact (computed by code)
F56.E1	Predicted class: Generic with calibrated probability above 0.99
F56.E2	Conformal prediction set (90% coverage): Generic
F56.E3	Column service = dns supports Generic (contribution +6.163)
F56.E4	Column fe_total_bytes = 114 supports Generic (contribution +0.644)
F56.E5	Column smean = 57 supports Generic (contribution +0.538)
F56.E6	Column ct_dst_sport_ltm = 25 supports Generic (contribution +0.538)
F56.E7	Column ct_dst_ltm = 25 supports Generic (contribution +0.529)
F56.E8	In the analysed dataset, 30 flows (11.3% of the predicted attacks) were classified as Generic, with average confidence 0.98

LLM explanation (passed checks). The model is 0.99 confident that this flow is Generic, and its 90% prediction set contains Generic. The strongest evidence is service = dns (contribution +6.163), fe_total_bytes = 114 (contribution +0.644), and smean = 57 (contribution +0.538), while ct_dst_sport_ltm = 25 and ct_dst_ltm = 25 also support Generic (contribution +0.538 and +0.529, respectively). In the analysed dataset, 30 flows (11.3% of the predicted attacks) were classified as Generic, with average confidence 0.98. [F56.E1] [F56.E2] [F56.E3] [F56.E4] [F56.E5] [F56.E6] [F56.E7] [F56.E8]

Generation: status generated, model qwen2.5:7b, 8 s, 5 statements checked, 0 rejected.

F84 (Fuzzers)

Evidence id	Fact (computed by code)
F84.E1	Predicted class: Fuzzers with calibrated probability 0.72
F84.E2	Conformal prediction set (90% coverage): Fuzzers
F84.E3	Column sbytes = 774 supports Fuzzers (contribution +1.487)
F84.E4	Column sttl = 254 supports Fuzzers (contribution +1.032)
F84.E5	Column dbytes = 642 supports Fuzzers (contribution +0.612)
F84.E6	Column service = - supports Fuzzers (contribution +0.407)
F84.E7	Column synack = 0.0678 supports Fuzzers (contribution +0.251)
F84.E8	In the analysed dataset, 27 flows (10.2% of the predicted attacks) were classified as Fuzzers, with average confidence 0.71

LLM explanation (passed checks). The model is 0.72 confident that this flow is Fuzzers with the 90% prediction set containing Fuzzers. The strongest evidence is sbytes = 774 (contribution +1.487) and sttl = 254 (contribution +1.032), while dbytes = 642 (contribution +0.612) and synack = 0.0678 (contribution +0.251) support it. In the analysed dataset, 27 flows (10.2% of the predicted attacks) were classified as Fuzzers, with average confidence 0.71. [F84.E1] [F84.E2] [F84.E3] [F84.E4] [F84.E5] [F84.E6] [F84.E7] [F84.E8]

Generation: status generated, model qwen2.5:7b, 7 s, 4 statements checked, 0 rejected.

F172 (Worms)

Evidence id	Fact (computed by code)
F172.E1	Predicted class: Worms with calibrated probability 0.99
F172.E2	Conformal prediction set (90% coverage): Worms
F172.E3	Column service = http supports Worms (contribution +3.473)
F172.E4	Column sttl = 254 supports Worms (contribution +3.149)
F172.E5	Column sbytes = 2416 supports Worms (contribution +3.009)
F172.E6	Column smean = 50 supports Worms (contribution +1.774)
F172.E7	Column fe_byte_balance = -0.9845 supports Worms (contribution +1.567)
F172.E8	In the analysed dataset, 18 flows (6.8% of the predicted attacks) were classified as Worms, with average confidence 0.78

LLM explanation (passed checks). The model is 0.99 confident that this flow is Worms with the 90% prediction set containing Worms. The strongest evidence is service = http (contribution +3.473) and sttl = 254 (contribution +3.149), while sbytes = 2416 (contribution +3.009) and smean = 50 (contribution +1.774) support it. In the analysed dataset, 18 flows (6.8% of the predicted attacks) were classified as Worms, with average confidence 0.78. [F172.E1] [F172.E2] [F172.E3] [F172.E4] [F172.E5] [F172.E6] [F172.E7] [F172.E8]

Generation: status generated, model qwen2.5:7b, 7 s, 5 statements checked, 0 rejected.

Appendix C. Sources

Source documents

Source	Publisher and title	Version / dates	Document link and hash	Cited
CISA IR Playbooks	CISA Cybersecurity Incident & Vulnerability Response Playbooks	- PDF created 2024-08-23 retrieved 2026-10-18	download sha256 2277247542d844d2258c...	7
MITRE ATT&CK	The MITRE Corporation MITRE ATT&CK, Enterprise matrix (STIX 2 bundle)	matrix object modified 2026-07-31; newest technique modified 2026-10-12 retrieved 2026-10-18	download sha256 66f3fe3f07ef1a2de0bd...	10
MITRE ATT&CK mitigation	The MITRE Corporation MITRE ATT&CK, Enterprise matrix (STIX 2 bundle)	matrix object modified 2026-07-31; newest technique modified 2026-10-12 retrieved 2026-10-18	download sha256 66f3fe3f07ef1a2de0bd...	12
NIST SP 800-61r3	NIST Incident Response Recommendations and Considerations for Cybersecurity Risk Management: A CSF 2.0 Community Profile	SP 800-61 Revision 3 PDF created 2025-04-01 retrieved 2026-10-18	download sha256 e5593d6bb85daecec7e8...	1
NIST SP 800-94	NIST NIST SP 800-94, Guide to Intrusion Detection and Prevention Systems (IDPS)	SP 800-94 PDF created 2007-02-15 retrieved 2026-10-18	download sha256 289c2ed27f7e3aac4162...	2

Source register (passages cited in this report)

Id	Source and location	Retrieval	Passage (excerpt; full text in the web view and JSON)
S1	MITRE ATT&CK T1498 sha256 fe2ae28205dada5a...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1498 Network Denial of Service. Tactics: impact. Adversaries may perform Network Denial of Service (DoS) attacks to degrade or block the availability of targeted resources to users. Network DoS can be performed by exhausting the network bandwidth services rely on. Example resources include specific websites, email services, DNS, and web-based applications. ...

Id	Source and location	Retrieval	Passage (excerpt; full text in the web view and JSON)
S2	MITRE ATT&CK T1499 sha256 12926c3ef5633f0d...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1499 Endpoint Denial of Service. Tactics: impact. Adversaries may perform Endpoint Denial of Service (DoS) attacks to degrade or block the availability of services to users. Endpoint DoS can be performed by exhausting the system resources those services are hosted on or exploiting the system to cause a persistent crash condition. ...
S3	MITRE ATT&CK T1046 sha256 a6753f58bd7da7a9...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1046 Network Service Discovery. Tactics: discovery. Adversaries may attempt to get a listing of services running on remote hosts and local network infrastructure devices, including those that may be vulnerable to remote software exploitation. ...
S4	MITRE ATT&CK T1595 sha256 84a571b28f37c796...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1595 Active Scanning. Tactics: reconnaissance. Adversaries may execute active reconnaissance scans to gather information that can be used during targeting. Active scans are those where the adversary probes victim infrastructure via network traffic, as opposed to other forms of reconnaissance that do not involve direct interaction. ...
S5	MITRE ATT&CK mitigation M1056 Pre-compromise (for T1595) sha256 e00239a46a941968...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Pre-compromise: This technique cannot be easily mitigated with preventive controls since it is based on behaviors performed outside of the scope of enterprise defenses and controls. Efforts should focus on minimizing the amount and sensitivity of data available to external parties.
S6	MITRE ATT&CK mitigation M1042 Disable or Remove Feature or Program (for T1046) sha256 fc4806cb66e13d45...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Disable or Remove Feature or Program: Ensure that unnecessary ports and services are closed to prevent risk of discovery and potential exploitation.
S7	MITRE ATT&CK mitigation M1037 Filter Network Traffic (for T1498) sha256 9a1c430b2caf8c55...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Filter Network Traffic: When flood volumes exceed the capacity of the network connection being targeted, it is typically necessary to intercept the incoming traffic upstream to filter out the attack traffic from the legitimate traffic. ...
S8	MITRE ATT&CK mitigation M1031 Network Intrusion Prevention (for T1046) sha256 c9135a904bd1a390...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Network Intrusion Prevention: Use network intrusion detection/prevention systems to detect and prevent remote service scans.
S9	NIST SP 800-94 page 68 sha256 b5b479ad633d5f3e...	TF-IDF similarity to the class and technique names; score 0.1021	These attacks typically involve significantly increased bandwidth usage or a much larger number of packets or connections to or from a particular host than usual. By monitoring these characteristics, anomaly detection methods can determine if the observed activity is significantly different than the expected activity. ...
S10	NIST SP 800-94 page 59 sha256 3b0e92227cdc5f2c...	TF-IDF similarity to the class and technique names; score 0.1001	They cannot detect the use of passive sensors that simply monitor and analyze observed traffic.38 Denial of service (DoS) attacks and conditions (e.g., network interference). ...
S11	MITRE ATT&CK T1190 sha256 d3a700fe87e6d49c...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1190 Exploit Public-Facing Application. Tactics: initial-access. Adversaries may attempt to exploit a weakness in an Internet-facing host or system to initially access a network. The weakness in the system can be a software bug, a temporary glitch, or a misconfiguration. ...
S12	MITRE ATT&CK T1210 sha256 b79a833daefa4ece...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1210 Exploitation of Remote Services. Tactics: lateral-movement. Adversaries may exploit remote services to gain unauthorized access to internal systems once inside of a network. ...
S13	MITRE ATT&CK mitigation M1042 Disable or Remove Feature or Program (for T1210) sha256 f018d527bce7c9b3...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Disable or Remove Feature or Program: Minimize available services to only those that are necessary.
S14	MITRE ATT&CK mitigation M1048 Application Isolation and Sandboxing (for T1190) sha256 b46ba7e2a134f815...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Application Isolation and Sandboxing: Application isolation will limit what other processes and system features the exploited target can access.
S15	MITRE ATT&CK mitigation M1016 Vulnerability Scanning (for T1210) sha256 a0184902520e34b2...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Vulnerability Scanning: Regularly scan the internal network for available services to identify new and potentially vulnerable services.

Id	Source and location	Retrieval	Passage (excerpt; full text in the web view and JSON)
S16	MITRE ATT&CK mitigation M1037 Filter Network Traffic (for T1190) sha256 caf776bda7df18de...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Filter Network Traffic: Restrict outbound network traffic from public-facing servers to prevent unauthorized connections from initiating communications with attacker-controlled infrastructure. ...
S17	CISA IR Playbooks page 12 sha256 69a6602c3ec22323...	TF-IDF similarity to the class and technique names; score 0.1603	Table 1: Example Adversary Tactics, Techniques, and Relevant Log and Event Data Tactic Common Techniques Log and Event Sources Indicators Initial Access Phishing [T1566], Drive-by Compromise [T1189], Exploit Public Facing Application [T1190], External Remote Services [T1133] Email, web proxy, server application logs, IDS/IPS Phishing, redirect, and payload ...
S18	CISA IR Playbooks page 31 sha256 027e7cbc3437f766...	TF-IDF similarity to the class and technique names; score 0.1053	TLP:CLEAR Step Incident Response Procedure Action Taken Date Completed 7c. Coordinate with law enforcement to collect and preserve evidence (as required by (Step 3a) prior to eradication, if applicable. 7d. ...
S19	MITRE ATT&CK T1600 sha256 f848c7fb14770438...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1600 Weaken Encryption. Tactics: defense-impairment. Adversaries may compromise a network device’s encryption capability in order to bypass encryption that would otherwise protect data communications. ...
S20	CISA IR Playbooks page 31 sha256 8ea4c248cca62261...	TF-IDF similarity to the class and technique names; score 0.1022	7k. Monitor for signs of threat actor response to containment activities. 7l. Report updated timeline and findings (including new atomic and behavioral indicators) to CISA. 7m. If new signs of compromise are found, return to technical analysis (Step 4) to re-scope the incident. 7n. ...
S21	MITRE ATT&CK T1595.002 sha256 c5c97b4f3b640ef2...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1595.002 Vulnerability Scanning. Tactics: reconnaissance. Adversaries may scan victims for vulnerabilities that can be used during targeting. Vulnerability scans typically check if the configuration of a target host/application (ex: software and version) potentially aligns with the target of a specific exploit the adversary may seek to use. ...
S22	MITRE ATT&CK T1499.004 sha256 dfb90bafb149428b...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1499.004 Application or System Exploitation. Tactics: impact. Adversaries may exploit software vulnerabilities that can cause an application or system to crash and deny availability to users. ...
S23	MITRE ATT&CK mitigation M1037 Filter Network Traffic (for T1499.004) sha256 1e822ce891cbf5f4...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Filter Network Traffic: Leverage services provided by Content Delivery Networks (CDN) or providers specializing in DoS mitigations to filter traffic upstream from services. Filter boundary traffic by blocking source addresses sourcing the attack, blocking ports that are being targeted, or blocking protocols being used for transport.
S24	MITRE ATT&CK mitigation M1056 Pre-compromise (for T1595.002) sha256 e00239a46a941968...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Pre-compromise: This technique cannot be easily mitigated with preventive controls since it is based on behaviors performed outside of the scope of enterprise defenses and controls. Efforts should focus on minimizing the amount and sensitivity of data available to external parties.
S25	CISA IR Playbooks page 23 sha256 2883387114bc8170...	TF-IDF similarity to the class and technique names; score 0.1248	The system is not vulnerable. - Susceptible. The system is vulnerable, but no signs of exploitation were found, and remediation has begun. - Compromised. The system was vulnerable, signs of exploitation were found, and incident response and vulnerability remediation has begun. ...
S26	CISA IR Playbooks page 23 sha256 d693aa3b2b7962a4...	TF-IDF similarity to the class and technique names; score 0.099	If the vulnerability exists in the environment, address the vulnerability itself—as described in the Remediation section below—and determine whether it has been exploited in the agency's environment. Use existing best practices to find signs of exploitation, including: - A sweep for known IOCs associated with exploitation of the vulnerability. ...
S27	MITRE ATT&CK T1570 sha256 d4a18c577a354d76...	ATT&CK technique mapped to the class (fixed mapping, by ID)	T1570 Lateral Tool Transfer. Tactics: lateral-movement. Adversaries may transfer tools or other files between systems in a compromised environment. ...
S28	MITRE ATT&CK mitigation M1050 Exploit Protection (for T1210) sha256 874e479adc4236b9...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Exploit Protection: Security applications that look for behavior used during exploitation such as Windows Defender Exploit Guard (WDEG) and the Enhanced Mitigation Experience Toolkit (EMET) can be used to mitigate some exploitation behavior. Control flow integrity checking is another way to potentially identify and stop a software exploit from occurring. ...
S29	MITRE ATT&CK mitigation M1030 Network Segmentation (for T1210) sha256 b4812184a1e4cfeb...	ATT&CK mitigation linked to a mapped technique (STIX "mitigates")	Network Segmentation: Segment networks and systems appropriately to reduce access to critical systems and services to controlled methods.

Id	Source and location	Retrieval	Passage (excerpt; full text in the web view and JSON)
S30	CISA IR Playbooks page 12 sha256 70801a6c33cf13ff...	TF-IDF similarity to the class and technique names; score 0.0812	Lateral Movement Exploitation of Remote Services [T1210], Remote Session Hijacking [T1563], Software Deployment Tools [T1072] Internal network logs, host event logs, Application Logs Mismatch of users and applications/credentials, workstation to workstation communication, beaconing from hosts not intended to be internet accessible, etc. ...
S31	NIST SP 800-61r3 page 36 sha256 ce7f9c9d56dae692...	TF-IDF similarity to a response-phase query; score 0.1051	R1: Track and validate the status of all ongoing incidents so that incidents in need of more response resources or a change in response strategy can be identified and the necessary changes initiated rapidly. ...
S32	CISA IR Playbooks page 15 sha256 e76b16c25038117a...	TF-IDF similarity to a response-phase query; score 0.186	See NIST SP 800-184: Guide for Cybersecurity Event Recovery for additional guidance. - Developing response scenarios for threat actor use of alternative attack vectors. - Allowing adequate time to ensure all systems are clear of all possible threat actor persistence mechanisms (backdoors, etc.) as adversaries often use more than one mechanism. ...

Scraping/ingestion collected these documents; retrieval selected passages; the LLM may only cite them. A retrieval score measures word similarity, not the probability that a claim is true.

Appendix D. Provenance and technical metadata

Item	Value
Detector	LightGBM v3 (class offsets, chosen without test labels) (v3); bundle built 2026-10-26T14:48:53+00:00 from commit be09e74cca
model.txt	8ff9a303ef36d60f54a15ee9850afa43ed2f7f6eb4015a32874e737f01e9ad83
preprocessing.json	de04362be907e9d7e292d0283757caa72b65b949900396876dbed84e3d127d42
calibration.json	d2a5c2658b69d8385e41b2f4977935db0d65291ac1dc9734ac8b2b8f08b1740e
model_card.json	18177a43efd705fe8a6785f16781d9b93ef3f4b99386cc667bd336586912cd4e
Preprocessing	42 UNSW-NB15 flow features + fe_total_bytes, fe_byte_balance, encoded into 195 model columns
Calibration	temperature 1.159; class offsets [-0.9, -0.9, 0.8, 0.0, -0.2, 0.2, 0.0, -0.4, -0.8, 0.0]; alpha 0.1
Explanation	TreeSHAP (LightGBM pred_contrib) of the base booster, grouped by original column; exact for the deployed class score: (base value + contributions) / temperature + class offset; not a decomposition of the probability
Knowledge base	downloaded 2026-10-18; chunks sha256 73a33392c414a28bf26bcf55e37883ad59813380dbe93220a62e34e79670f2a5
LLM	qwen2.5:7b (local Ollama), temperature 0, seed 42; uploaded flows are sent only to the local model
Input file sha256	e8f8eb325c36dd3c83e88c0d9a9aebce006d49f6ad67887c873aa7c327e61700
Application	code commit e4bf7aed6ab9d00bc56a4f7021872d0cad9f73ef; report schema 2.1
Timings (s)	detections_summary 0.038, group_xai 3.538, xai 0.867, retrieval 0.132, generation 37.804, checks 0.027, total_record 42.517

Appendix E. Model benchmark (official test file; not measured on this upload)

Benchmark results on the official UNSW-NB15 test file: context for the model, not performance measured on this upload.

Metric	Attack vs normal	10-class macro	10-class weighted
Accuracy	86.5%	74.5%	74.5%
Precision	81.5%	53.8%	84.2%
Recall	97.6%	58.6%	74.5%
F1	88.8%	54.1%	77.9%

Best possible on that file (a model that already knows the labels): accuracy 93.6%, macro F1 76.5%. Top-2 accuracy 91.2%. Normal flows flagged: 27.2%.

Confusion matrix (official test file)

True class (rows) / predicted (columns)	1	2	3	4	5	6	7	8	9	10
1. Analysis	50	71	500	40	0	0	16	0	0	0
2. Backdoor	50	45	462	19	1	0	1	0	5	0
3. DoS	146	867	2,427	516	43	8	20	29	32	1
4. Exploits	185	804	2,620	7,063	110	27	71	169	79	4
5. Fuzzers	112	132	1,059	274	3,333	4	946	5	197	0
6. Generic	0	5	140	354	26	18,317	5	0	19	5
7. Normal	482	1	227	654	8,594	7	26,937	5	93	0
8. Reconnaissance	13	114	231	273	12	1	11	2,808	30	3
9. Shellcode	0	0	21	21	21	0	5	1	308	1
10. Worms	0	0	1	16	0	0	0	0	0	27

Columns use the same numbers as the rows: 1 = Analysis; 2 = Backdoor; 3 = DoS; 4 = Exploits; 5 = Fuzzers; 6 = Generic; 7 = Normal; 8 = Reconnaissance; 9 = Shellcode; 10 = Worms. Green diagonal = correct; red = mistakes (darker = larger share of that true class).

Appendix F. Comparison with the labels in the uploaded file

Labels found in the uploaded file were compared with the predictions after the analysis. They were not used for predictions, explanations, retrieval or actions.

Check	Result
Attacks predicted as attacks	261 of 270 (96.7%)
Normal flows predicted as attacks	5 of 30 (16.7%)
Exact class right	177 of 300 (59.0%)
True class on the 90% shortlist	244 of 300